THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — MATHS · HARD LEVEL MCQ TEST SET 3

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 1 Hour Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. Hard level: application and higher-order questions.

MULTIPLE CHOICE QUESTIONS — 50 × 1 = 50

Choose the correct option and write it in your answer book.

1. The pair $kx + 7y = k - 7$ and $28x + ky = k$ has infinitely many solutions when $k =$ 1

(a) 14 **(b)** -14 **(c)** 28 **(d)** 7

2. If the roots of $x^2 - 12x + k = 0$ differ by 4, then $k =$ 1

(a) 64 **(b)** 128 **(c)** 32 **(d)** 40

3. The sum of the first 9 terms of an AP is 279, and its first term is 3. The 9th term is: 1

(a) 28 **(b)** 65 **(c)** 62 **(d)** 59

4. In a right triangle with hypotenuse $c = 13$ and one leg $a = 5$, the area (sq. units) is: 1

(a) 60 **(b)** 30 **(c)** $\frac{65}{2}$ **(d)** 78

5. The area of the triangle formed by $(11, 0)$, $(0, 4)$ and the origin, doubled, is: 1

(a) 88 **(b)** 44 **(c)** 22 **(d)** 15

6. If $\tan\theta = \frac{12}{5}$, then $\dfrac{1 - \tan^2\theta}{1 + \tan^2\theta} =$ 1

(a) $\frac{5}{12}$ **(b)** $-\frac{119}{169}$ **(c)** $\frac{120}{169}$ **(d)** $\frac{119}{169}$

7. The angle of elevation of the top of a vertical tower from a point on level ground is $30°$. After walking 6 m towards the foot of the tower along a straight line, the angle of elevation becomes $45°$. The height of the tower is: 1

(a) $2\sqrt{3}$ m **(b)** 6 m **(c)** $3 + 3\sqrt{3}$ m **(d)** $9 + 3\sqrt{3}$ m

8. Tangents PA and PB are drawn from a point P, 50 cm from the centre O of a circle of radius 14 cm. The length of each tangent is 48 cm. The length of the chord AB is: 1

(a) 28 cm **(b)** 13.44 cm **(c)** 96 cm **(d)** 26.88 cm

9. A chord of a circle of radius 10.5 cm subtends an angle of $120°$ at the centre. (Use $\pi = \frac{22}{7}$ and $\sqrt{3} = 1.73$; answer to two decimal places.) The area of the minor segment is: 1

(a) 115.50 cm² **(b)** 67.82 cm² **(c)** 163.18 cm² **(d)** 47.68 cm²

10. A hemispherical bowl has outer radius 9 cm and inner radius 7.5 cm. The volume of material used to make the bowl is: 1

(a) 1287 cm³ **(b)** 643.5 cm³ **(c)** 1930.5 cm³ **(d)** 321.75 cm³

11. The median of a distribution with classes 0–10, 10–20, 20–30, 30–40, 40–50, 50–60, 60–70 and frequencies 20, x, 18, y, 12, 7, 18 respectively, and total frequency 100, is 25. The value of x is: 1

(a) 22 **(b)** 6 **(c)** 21 **(d)** 20

12. Two fair dice are thrown together and the numbers on the upper faces are noted. The probability that the sum of the numbers is 7 is: 1

(a) $\frac{7}{36}$ **(b)** $\frac{5}{36}$ **(c)** $\frac{1}{6}$ **(d)** $\frac{5}{6}$

13. The largest number dividing 75, 123 and 195 leaving remainder 3 in each case is: 1

(a) 6 **(b)** 12 **(c)** 48 **(d)** 24

14. If α, β are zeroes of $x^2 - 8x + 7$, then $\alpha^2\beta + \alpha\beta^2$ is: 1

(a) 64 **(b)** 15 **(c)** 56 **(d)** 49

15. If $13x + 5y = 67$ and $6x + 7y = 45$, then $x + y =$ 1

(a) 3 **(b)** 4 **(c)** 7 **(d)** 12

16. For which value of k does $x^2 + kx + 16 = 0$ have equal roots (take $k > 0$)? 1

(a) 4 **(b)** 16 **(c)** 32 **(d)** 8

17. In an AP, $S_{10} = 315$ and $S_9 = 261$. Then $a_{10} =$ 1

(a) 49 **(b)** 59 **(c)** 54 **(d)** 576

18. The altitude on the hypotenuse of the right triangle with legs 5 and 12 is of length (cm): 1

(a) 30 **(b)** $\frac{13}{2}$ **(c)** $\frac{17}{2}$ **(d)** $\frac{60}{13}$

19. If $P(x, y)$ is equidistant from $(2, 4)$ and $(8, 2)$ and lies on $y = 0$, then $x =$ 1

(a) 8 **(b)** 5 **(c)** 4 **(d)** -4

20. If $\sin\theta = \frac{12}{13}$, the value of $2\sin\theta\cos\theta$ is: 1

(a) $\frac{120}{169}$ **(b)** $\frac{10}{13}$ **(c)** $\frac{60}{169}$ **(d)** $\frac{24}{13}$

21. A flagstaff stands on the top of a building 12 m high. From a point on level ground the angles of elevation of the bottom and the top of the flagstaff are 30° and 60° respectively. The length of the flagstaff is: 1

(a) 36 m **(b)** 24 m **(c)** $24\sqrt{3}$ m **(d)** 48 m

22. PA and PB are tangents from P to a circle, each of length 11 cm. A third tangent touches the circle at E on the minor arc AB and meets PA at C and PB at D. The perimeter of $\triangle PCD$ is: 1

(a) 13 cm **(b)** 11 cm **(c)** 22 cm **(d)** 33 cm

23. A brooch is a circular piece of silver wire of diameter 49 mm. The wire is also used for 7 equal diameters that divide the brooch into 14 sectors. The total length of wire required (in mm) is: 1

(a) 325.5 mm **(b)** 1886.5 mm **(c)** 154 mm **(d)** 497 mm

24. The largest possible solid sphere is carved out of a cube of edge 42 cm. The volume of the material wasted is: 1

(a) 54684 cm³ **(b)** 38808 cm³ **(c)** 35280 cm³ **(d)** 74088 cm³

25. For the data with classes 40–50, 50–60, 60–70, 70–80, 80–90, 90–100, 100–110, 110–120 with frequencies 21, 13, 25, 5, 6, 24, 4, 2 respectively, The mean of the data is: 1

(a) 85 **(b)** 71 **(c)** 72 **(d)** 887.5

26. Two fair dice are thrown together and the numbers on the upper faces are noted. The probability that the numbers differ by 2 is: 1

(a) $\frac{1}{4}$ **(b)** $\frac{2}{9}$ **(c)** $\frac{7}{36}$ **(d)** $\frac{7}{9}$

27. The smallest positive integer by which 18 must be multiplied to obtain a perfect square is: 1

(a) 3 **(b)** 2 **(c)** 6 **(d)** 18

28. If α, β are zeroes of $x^2 - 9x + 8$, then $|\alpha - \beta|$ is: 1

(a) 8 **(b)** 14 **(c)** 9 **(d)** 7

29. The area (in sq. units) of the triangle formed by the line $3x + 4y = 12$ and the coordinate axes is: 1

(a) 2 **(b)** 6 **(c)** 12 **(d)** 72

30. If $x = 2$ is a root of $kx^2 - 3x - 2 = 0$, then $k =$ 1

(a) -2 **(b)** 2 **(c)** 1 **(d)** $\frac{1}{2}$

31. The sum of the first n terms of an AP is $S_n = 3n^2 + 1n$. Its common difference is: 1

(a) 3 **(b)** 6 **(c)** 4 **(d)** 1

32. If $\triangle ABC \sim \triangle PQR$, $ar(ABC) = 48$ and $ar(PQR) = 243$ and $AB = 8$, then $PQ =$ 1

(a) 22 **(b)** 8 **(c)** 162 **(d)** 18

33. The ratio in which $(-2, 4)$ and $(8, 1)$ are joined by the y-axis is 2: 1

(a) 8 **(b)** 1 **(c)** 6 **(d)** 2

34. If $\sec\theta + \tan\theta = 5$, then $\sec\theta - \tan\theta =$ 1

(a) 5 **(b)** 1 **(c)** $\frac{1}{5}$ **(d)** 2

35. A car moves at a uniform speed towards the foot of a vertical tower. The angle of elevation of the top of the tower changes from 30° to 60° in 8 minutes. The further time the car takes to reach the foot of the tower is: 1

(a) 16 minutes **(b)** $\frac{8}{3}$ minutes **(c)** 8 minutes **(d)** 4 minutes

36. Tangents QA and QB are drawn from an external point Q to a circle, and $QA = (12x + 7)$ cm and $QB = (5x + 28)$ cm. The length of QA is: 1

(a) 64 cm **(b)** 45 cm **(c)** 3 cm **(d)** 43 cm

37. A horse is tied at the midpoint of one side of a large square field with a rope 42 m long. The area of the field it can graze is: 1

(a) 5544 m² **(b)** 66 m² **(c)** 132 m² **(d)** 2772 m²

38. A solid sphere of radius 21 cm fits exactly inside a cylinder (it touches the curved surface and both bases). The volume of the space in the cylinder not occupied by the sphere is: 1

(a) 38808 cm³ **(b)** 19404 cm³ **(c)** 58212 cm³ **(d)** 97020 cm³

39. For the data with classes 100–120, 120–140, 140–160, 160–180, 180–200, 200–220, 220–240, 240–260 with frequencies 4, 12, 2, 21, 22, 21, 14, 4 respectively, The median of the data is: 1

(a) 200 **(b)** 180 **(c)** 190 **(d)** 186.8

40. A bag contains 11 red balls and x blue balls. The probability of drawing a blue ball is 4 times the probability of drawing a red ball. The value of x is: 1

(a) 4 **(b)** 43 **(c)** 44 **(d)** 55

41. **Assertion (A):** The mid-point of the segment joining $(2, -3)$ and $(8, 5)$ is $(5, 1)$. 1

Reason (R): The mid-point of the segment joining (x_1, y_1) and (x_2, y_2) is $\left(\frac{x_1 - x_2}{2}, \frac{y_1 - y_2}{2}\right)$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

42. **Assertion (A):** $\tan 45^\circ = 1$. 1

Reason (R): $\tan\theta = \frac{\cos\theta}{\sin\theta}$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

43. **Assertion (A):** A tower $10\sqrt{3}$ m high casts a shadow 10 m long when the angle of elevation of the Sun is 60°. 1

Reason (R): $\tan 60^\circ = \frac{1}{\sqrt{3}}$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

44. **Assertion (A):** Exactly two tangents can be drawn to a circle from a point outside it. 1

Reason (R): A tangent to a circle intersects the circle at two points.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

45. **Assertion (A):** The length of the arc of a sector of radius 21 cm and angle 60° is 22 cm (use $\pi = \frac{22}{7}$). 1

Reason (R): The length of the arc of a sector of angle θ is $\frac{\theta}{360^\circ} \times \pi r^2$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

46. **Assertion (A):** A cone and a cylinder have the same base radius and the same height. The volume of the cone is one-third that of the cylinder. 1

Reason (R): The volume of a cone of base radius r and height h is $\pi r^2 h$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

47. **Assertion (A):** For grouped data the median class is the class whose cumulative frequency is just greater than or equal to $\frac{n}{2}$. 1

Reason (R): The median class is the class which has the highest frequency.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

48. **Assertion (A):** The probability of getting a prime number when a die is thrown once is $\frac{1}{2}$. 1

Reason (R): The probability of an event can be greater than 1.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

49. **Assertion (A):** $\sqrt{2} \times \sqrt{8}$ is a rational number. 1

Reason (R): The product of two irrational numbers is always irrational.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

50. **Assertion (A):** The polynomial $x^2 + 1$ has two real zeros. 1

Reason (R): A quadratic polynomial can have at most two zeros.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

END OF QUESTION PAPER **Total: 50 Marks**